Hints: Euclidean geometry – 2. Conics, areas and projective maps

Class work

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. Polar equation: put F at the origin and let $d = d(F, l)$. A point at polar coordinates (r, φ) is at distance $d - r\cos\varphi$ from l, so $r = e(d - r\cos\varphi)$, i.e. $r = \frac{p}{1+e\cos\varphi}$. Canonical form: write $|XF|^2 = e^2 d(X,l)^2$ in Cartesian coordinates with the axis along the perpendicular to l and complete the square in x; for $e \neq 1$ the centre appears, and $c = ea$, directrices $x = \pm\frac{a}{e}$.

Hint 2. Focal sum: $|XF_1| + |XF_2| = e\big(d(X,l_1) + d(X,l_2)\big) = e \cdot \frac{2a}{e} = 2a$, since X lies between the two directrices. Reflection: every point $Y \neq X$ of the tangent at X lies outside the ellipse, so $|YF_1| + |YF_2| > 2a = |XF_1| + |XF_2|$; by Heron's reflection principle the tangent makes equal angles with XF_1 and XF_2. For the hyperbola and the parabola argue in the same way with $\big||YF_1| - |YF_2|\big|$ and with $|YF| - d(Y,l)$.

Hint 3. A ray through F_1 is reflected into a ray through F_2, and then again through F_1, and so on. For Fact 1 write the normal at $(t, \frac{t^2}{2p})$: $(x - t) + \frac{t}{p}(y - \frac{t^2}{2p}) = 0$; substituting (u, v) and multiplying by $-2p^2$ gives the cubic. It has no t^2 term, so the roots add up to 0, and it has three real roots iff its discriminant $-4P^3 - 27Q^2$ ($P = 2p(p - v)$, $Q = -2p^2u$) is positive, i.e. $8(v - p)^3 > 27pu^2$.

Problem 2

Hint 1. The determinant formula is the area of the parallelogram spanned by $B - A$ and $C - A$, halved, with the sign of the orientation. The map $P \mapsto [PBC] + [PCA] + [PAB]$ is affine, and at $P = A$ it equals $[ABC]$ (one term, the others vanish); the same at B and C, so it is the constant $[ABC]$.

Hint 2. Cross-ratio: if h is the distance from O to the line, then $|XY| \cdot h = |OX|\,|OY| \sin\angle XOY$ (twice the area of OXY). Substituting into $\frac{(c-a)(d-b)}{(c-b)(d-a)}$ the lengths $|OA|, \ldots, |OD|$ and h cancel, which gives the formula with sines; it depends only on the four lines through O.

Hint 3. The value x is determined by $(a, b; c, x)$, so a cross-ratio preserving map is fixed by the images of three points, and a projective map is one. Fixed points of $x \mapsto \frac{\alpha x+\beta}{\gamma x+\delta}$ solve $\gamma x^2 + (\delta - \alpha)x - \beta = 0$ (check ∞ separately): at most two roots unless all coefficients vanish, i.e. the map is the identity.

Hint 4. Five points: five linear conditions on six coefficients have a non-zero solution. Uniqueness: if three of the points lie on a line L, every conic through them contains L, so it is L together with the unique line through the other two points. Otherwise, if $Q_1 \not\sim Q_2$ both pass through the points, choose $\lambda Q_1 + \mu Q_2 \neq 0$ vanishing at a further point of the line P_1P_2; it contains that line and therefore splits as P_1P_2 times a line through P_3, P_4, P_5, a contradiction.

Problem 3

Hint 1. Let F be the common focus. For the k-th ellipse $|XF| = e_k\, d(X, l_k)$, and the whole ellipse lies on one side of its directrix l_k, where $d(X, l_k)$ is an affine function of X.

Hint 2. So every common point satisfies $e_1 d(X, l_1) = |XF| = e_2 d(X, l_2)$, an affine equation: the common points lie on a line, and a line meets an ellipse at most twice. (If the two affine functions coincide, the equations $|XF| = e_k d(X, l_k)$ coincide and so do the ellipses.)

Problem 4

Hint 1. In the notation of Fact 1 the parabola is $x^2 = 2py$ with $p = \frac{1}{a}$. The normals through $A = (u, v)$ have feet with abscissas t solving $t^3 + 2p(p - v)t - 2p^2 u = 0$, and one of the roots is $t = u$ itself.

Hint 2. Factor out $t - u$ (for $u \neq 0$; the vertex $u = 0$ has no other normals through it): the other feet satisfy $t^2 + ut + 2p^2 = 0$.

Hint 3. B and B' are two distinct real roots, so $u^2 > 8p^2$, and $m = v = \frac{u^2}{2p} > 4p = \frac{4}{a}$.

Problem 5

Hint 1. Lemma: the midpoints of a family of parallel chords of a conic lie on a line, which passes through the centre for an ellipse or a hyperbola and is parallel to the axis for a parabola. Affine maps preserve midpoints, parallelism and centres, so it suffices to check it for a circle (a diameter) and for $y = x^2$ (chords of slope k have midpoints with $x = \frac{k}{2}$).

Hint 2. The parallel sides of the trapezoid are two parallel chords of each of the three conics. So the line joining their midpoints is the line of midpoints for all three conics at once.

Hint 3. Hence the centres of the ellipse and the hyperbola lie on this line, and the axis of the parabola is parallel to it.